PA4 - Polymorphism and Operator Overloading

CS200 - Spring 2024-2025

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1 Overview

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1.1 Plagiarism policy

The teaching staff will follow a strict no-plagiarism policy for programming assignments. If any case of plagiarism is observed, there will be a severe penalty. More extreme cases of plagiarism will be forwarded to the Disciplinary Committee.

- Copying code from the internet is strictly prohibited.
- The use of LLMs is not allowed for this assignment.
- Sharing code with other students, including pseudocode and flowcharts, is not allowed.
- You will be held accountable for sharing your code so it is your duty to prevent that.
- The course staff reserves the right to hold vivas at any point, during or after the semester.
- Any external assistance or discussion must be indicated at the time of submission (see below).

1.2 Submission instructions

- Zip the files `ACS_61.cpp` and `ACS_61.h` and use the following naming convention: `PA4_<rollnumber>.zip`
  For example, if your roll number is 28100042, your zip file should be: `PA4_28100042.zip`
- All submissions must be uploaded on the respective LMS assignment tab before the deadline.
  Any other forms of submission (email, dropbox, etc) will not be accepted.
- If you have had external assistance (such as sharing of code or use of LLMs) and you wish to tell us directly, you can include a "Collaboration.txt" file in your submission seperately, and explain in detail about the extent and source. We will still penalize accordingly if needed. However, doing so will ensure that your case will NOT end up being forwarded to the Disciplinary Committee.

1.3 Aims and learning outcomes

1.3.1 Aim

The aim of this programming project is to provide students with a thorough understanding of key Object-Oriented Programming (OOP) concepts i.e. inheritance, polymorphisn, and multiple inheritance in C++. Through the implementation of a custom Air Traffic Control (ATC) system, you will gain practical experience in problem-solving, understanding inter-class relationships, and overloading various types of operators.

1.3.2 Learning outcomes

- Design and implement a class hierarchy that effectively uses inheritance to model relationships between objects.
- Demonstrate multiple inheritance and address potential issues
- Utilize encapsulation to improve maintainability and scalability of your code
- Overload operators to extend class functionality in a natural and intuitive way.
- Understand how operator overloading can make classes more intuitive and easier to use.

1.4 Docker

Please note that your assignments will be tested on Docker containers. As such, it is recommended that you run your code on it at least once before submitting it. Should any errors arise on our end due to incompatibility, you will be given the chance to contest your code.

1.5 Grading distribution

Assignment breakdown	
Aircraft Class	(10 marks)
CombatAircraft Class	(10 marks)
StealthAircraft Class	(10 marks)
AbductorCraft Class	(10 marks)
GuardianCraft Class	(10 marks)
Operator Overloading	(50 marks)
Total	**100 marks**

1.6 Starter code

```
PA4
├─ Code
│  ├─ ACS_61.h
│  ├─ ACS_61.cpp
│  └─ Test_ACS_61.cpp
├─ Test
│  ├─ TestClasses
│  └─ TestOperators
└─ Manual
```

1.7 Code Quality

You are expected to follow good coding practices. We will manually look at your code and award points based on the following criteria:

- Readability: Use meaningful variable names as specified in the PEP 8 standards and lectures. Assure consistent indentation. We will be looking at the following:
  - **Variables, Functions, and Methods:** Use lowercase letters and separate words with underscores. For example, my_variable, calculate_sum()
  - **Constants:** Use uppercase letters and separate words with underscores. For example, MAX_SIZE, PI
  - **Classes:** Use CamelCase (capitalize the first letter of each word without underscores). For example, MyClass, CarModel
- **Modularity and Reusability:** Break down your code into functions with single responsibilities.

- **Code Readability:** You should make sure that your code is readable.
- **Documentation** (Optional): Comment your code adequately to explain the logic and flow.
- **Error Handling:** Appropriately handle possible errors and edge cases.
- **Efficiency:** Write code that performs well and avoids unnecessary computations.

1.8 Restrictions

Any violation of the below restrictions will result in very significant deductions.

- You are **not allowed to declare any other classes than those provided.** You are **not allowed to declare public-member attributes for any class.**
- You are **not allowed to declare any form of global or static variables**
- You are not allowed to include any other libraries other than those already provided in the starter code.
- You are not allowed to code in `ACS_61.h`, which is only for defining attributes and function signatures. **All code must go in `ACS_61.cpp`**.

For clarity, you are allowed to:

- You may declare helper functions.
- You may define class attributes and methods yourself to fit the functionality of that class.
- You may define inheritance relationships for classes in `ACS_61.h` yourself.

2 Assignment Overview

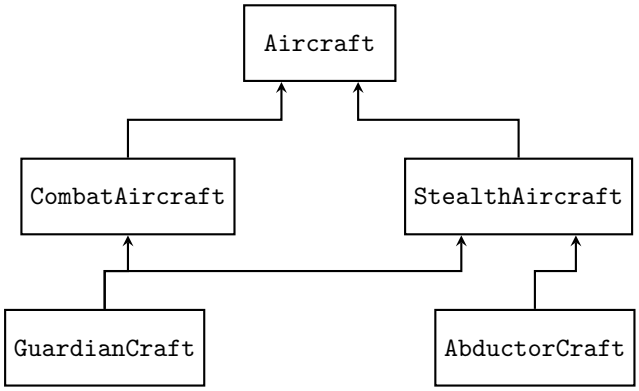
The clandestine Area 61 has been plagued by ongoing conflicts between villainous alien abductor aircraft, designed for stealth and human abductions, and Earth's guardian fleet, elite fighter jets equipped with cutting-edge cloaking and weaponry. To simulate potential engagement scenarios, the Base Commander of Area 61 has tasked you with replacing the outdated C-based simulator with a state-of-the-art C++ Air Combat Simulation (ACS) system. Here's what they have to say:

"These alien bandits have gone haywire – they're abducting people left, right, and center. The fate of our future generations depends on your CS-wizardry. My needs are simple: this object-oriented program must accurately model aircraft behavior, fuel management, stealth technology, and interactions between different aircraft types. Oh, and also, I need it by the 3rd of April!. Get to work."

3 Inheritance Implementation

You are to implement 5 classes: `Aircraft`, `CombatAircraft`, `StealthAircraft`, `AbductorCraft`, and `GuardianCraft` by following the inheritance structure below. For each class, you must include the attributes and function signatures in `ACS_61.h`. The code for all functions must go in `ACS_61.cpp`.

3.1 Inheritance Structure



An arrow pointing from a class to another class represents inheritance i.e. `AbductorCraft` inherits from `StealthAircraft`

3.2 General Requirements

In this assignment, you are not provided with method signatures. Instead, you are given set functionalities that must be met. You are **free to define methods and extra attributes according to your implementation** provided that the functionalities are met.

Outlined below are recommendations and general requirements that you must meet:

- **Test cases rely on working constructors and getters for all classes.**
- Test cases for constructors of all classes will indicate whether **essential constructor tests** have passed. Only these constructors are necessary pre-requisites for subsequent class functionalities and operator overloading tests.
- **All functionalities of ancestral classes must hold in inherited classes.**
- **All restrictions imposed in ancestral classes must hold in inherited classes.**
- You are free to define as many helper functions as you wish.
- Our final grading will be based on the provided test cases.

3.3 `Aircraft` Class (10 marks)

This class serves as the base class for all other classes.

3.3.1 Required Attributes

```
string identifier;
int fuel_level; // Stores current fuel level (0 to 100)
int health; // Stores current health (0 to 100)
AircraftStatus current_status;  // Stores current status (OnGround, Airborne, or Crashed) using given enum class
```

3.3.2 Initialization

You are to define constructors so that all of the following are valid ways to initialize `Aircraft` objects.

```
Aircraft a1;
Aircraft a2("ACFT2", 60, 100, AircraftStatus::OnGround); // i.e. Aircraft with 60% fuel, 100% health, on ground.
Aircraft a3(a2);
```

Notes:

- By default, aircraft are initialized on ground with full fuel and health.
- No aircraft can have at any point, or be initialized with, out-of-range health or fuel i.e. if an aircraft is initialized with -5 fuel, it will be automatically initialized with 0 fuel. Follow this line of thinking throughout this assignment.
- At any point, if an airborne aircraft runs out, or is initialized with 0, fuel or health, it crashes.
- All crashed aircraft have 0 health.
- No changes to an aircraft can be made after it has crashed.

3.3.3 Functionalities

You are to define methods for the following function calls:

```
string identifier = a2.getIdentifier();
int fuel_level = a2.getFuelLevel();
AircraftStatus current_status = a2.getCurrentStatus();

a2.takeOff();

a2.land();
```

Notes:

- For an aircraft to take off, it must have a minimum fuel of 30%.

3.4 `CombatAircraft` Class (10 marks)

This class is a subclass of the `Aircraft` class.

3.4.1 Required Attributes

```
string weapon_type;
int weapon_count; // Stores number of weapons on board
int weapon_strength; // Stores weapon strength (0 to 100)
```

3.4.2 Initialization

You are to define constructors so that all of the following are valid ways to initialize `CombatAircraft` objects.

```
CombatAircraft ca1;
CombatAircraft ca2("CMBT2", 60, 100, AircraftStatus::OnGround, "Missiles", 10, 50); // i.e. CombatAircraft with 10 missiles on board, weapon strength of 50.
CombatAircraft ca3(ca2);
```

Notes:

- By default, combat aircraft are initialized like default aircraft and with empty weapon type and no weapons on board.
- No combat aircraft can have at any point, or be initialized with, out-of-range weapon count or weapon strength.

3.4.3 Functionalities

You are to define methods for the following function calls:

```
string weapon_type = ca2.getWeaponType();
int weapon_count = ca2.getWeaponCount();
int weapon_strength = ca2.getWeaponStrength();
```

3.5 `StealthAircraft` Class (10 marks)

This class is a subclass of the `Aircraft` class.

3.5.1 Required Attributes

```
bool cloak_status; // Stores current status of stealth mechanism.
```

3.5.2 Initialization

You are to define constructors so that all of the following are valid ways to initialize `StealthAircraft` objects.

```
StealthAircraft sa1;
StealthAircraft sa2("STLH2", 60, 100, AircraftStatus::Airborne, false); // i.e. StealthAircraft with cloak off.
StealthAircraft sa3(sa2);
```

Notes:

- By default, stealth aircraft are initialized like default aircraft and with cloak off.
- No stealth aircraft can have at any point, or be initialized with, cloak on while it is not airborne.

3.5.3 Functionalities

You are to define methods for the following function calls:

```
bool cloak_status = sa2.getCloakStatus();

sa2.activateCloak();

sa2.deactivateCloak();
```

Notes:

- Activating cloak has a "starting" health cost of 10 and a "running" fuel cost of 15. Hint: think about what should happen if "starting" or "running" the cloak is too costly.
- There is no fuel or health cost when deactivating the cloak.
- If a stealth aircraft with its cloak on is simulated to land or take off, its cloak is first turned off.

3.6 `AbductorCraft` Class (10 marks)

This class is a subclass of the `StealthAircraft` class.

3.6.1 Required Attributes

```
int abductee_count; // Stores current number of abductees on board.
int abductee_capacity; // Stores maximum number of abductees.
```

3.6.2 Initialization

You are to define constructors so that all of the following are valid ways to initialize `AbductorCraft` objects.

```
AbductorCraft ac1;
AbductorCraft ac2("ABDT2", 60, 100, AircraftStatus::Airborne, true, 3, 10); // i.e. AbductorCraft with cloak on, 3 abductee on board with capacity of 10.
AbductorCraft ac3(ac2);
```

Notes:

- By default, abductor crafts are initialized like default stealth aircraft and with 0 abductee count and capacity.
- No abductor craft can have at any point, or be initialized with, out-of-range abductee count or abductee capacity.

3.6.3 Functionalities

You are to define methods for the following function calls:

```
int abductee_count = ac2.getAbducteeCount();
int abductee_capacity = ac2.getAbducteeCapacity();
```

3.7 `GuardianCraft` Class (10 marks)

This class is a subclass of both `CombatAircraft` and `StealthAircraft` classes.

3.7.1 Required Attributes

```
int kill_count; // Stores number of AbductorCraft's taken down.
```

3.7.2 Initialization

You are to define constructors so that all of the following are valid ways to initialize `GuardianCraft` objects.

```
GuardianCraft gc1;
GuardianCraft gc2("GRD2", 60, 100, AircraftStatus::Airborne, "Rockets", 6, 80, true, 5); // i.e. GuardianCraft with 6 rockets on board, weapon strength 80, cloak on, kill count = 5.
GuardianCraft gc3(gc2);
```

Notes:

- By default, guardian crafts are initialized like default combat and stealth aircraft and with 0 kill count.
- No guardian craft can have at any point, or be initialized with, out-of-range kill count.

3.7.3 Functionalities

You are to define methods for the following function calls:

```
int kill_count = gc2.getKillCount();
```

4 Operator Overloading Implementation

In this part of the assignment, you will overload operators to implement various simulations. You will code in the same files, remembering to only use `ACS_61.h` for function signatures. Code for these functions will go in `ACS_61.cpp`.

- It is up to you to determine valid return types and definitions of your overloaded operators based on the example usage and outputs.
- Operator overloading must be defined within classes, except where it is not possible.
- Overloaded operators for base classes must retain their usage in derived classes unless explicitly mentioned otherwise.
- The use of overloaded operators must not supersede previous restrictions outlined in the previous part of the assignment.

4.1 `AircraftStatus` Operator Overloading (2.5 marks)

4.1.1 Output Stream Operator «

Overload the output stream operator « in line with the following calls:

```
AircraftStatus status_ab = AicraftStatus::Airborne;
AircraftStatus status_on = AicraftStatus::OnGround;
AircraftStatus status_crashed = AicraftStatus::Crashed;

cout << status_ab << endl;
cout << status_on << endl;
cout << status_crashed << endl;
```

The terminal output of these calls should be exactly:

```
Airborne
On Ground
Crashed
```

4.2 `Aircraft` Operator Overloading

4.2.1 Refueling Operator +=

Overload the addition assignment operator `+=` to refuel Aircraft according to the following usage:

```
Aircraft a1("ACFT1", 34, 85, AircraftStatus::OnGround);

a1 += 20; // Adding 20% of fuel
```

Note: Aircraft can only be refueled when on ground.

4.2.2 Defueling Operator -=

Overload the subtraction assignment operator `+=` to defuel Aircraft according to the following usage:

```
a1 -= 30; // Removing 30% of fuel
```

Note: Aircraft can only be defueled when on ground.

4.2.3 Equality Operator ==

Overload the equality operator `==` to compare two aircraft objects according to the following usage:

```
Aircraft a2(a1);
if (a1 == a2) {
    cout << "Equal\n";
}
else {
    cout << "Not Equal\n";
}

Aircraft a4("ACFT4", 90, 84, AircraftStatus::Airborne);
if (a1 == a4) {
    cout << "Equal\n";
}
else {
    cout << "Not Equal\n";
}
```

The terminal output of these calls should be exactly:

```
Equal
Not Equal
```

Note: You may assume that this operator will be accessed only by `Aircraft` objects, not by objects of derived classes.

4.2.4 Output Stream Operator «

Overload the output stream operator « in line with calls of the following type:

```
cout << a1;
```

The terminal output of this call should be exactly:

```
Identifier: ACFT1
Fuel Level: 24%
Health: 85%
Current Status: On Ground
```

Note: Yes, there is an endline after the last line too.

4.2.5 Input Stream Operator »

Overload the input stream operator » in line with calls of the following type:

```
Aircraft a4;

cin >> a4;
```

This should prompt the user to enter attributes for the object. A terminal run of this call should be exactly:

```
Enter Identifier: FLY41
Enter Fuel Level (0-100): 59
Enter Health (0-100): 88
Enter Status (0: OnGround, 1: Airborne, 2: Crashed): 1
```

Note: You may assume that this operator will be accessed only by `Aircraft` objects, not by objects of derived classes.

4.3 `CombatAircraft` Operator Overloading

4.3.1 Weapon Post-Increment Operator ++

Overload the post-increment operator ++ to increment weapons according to the following usage:

```
CombatAircraft ca1("CMBT1", 89, 100, AircraftStatus::OnGround, "Missiles", 10, 40);

ca1++;
```

Note: Weapons can only be loaded when on ground.

4.3.2 Weapon Pre-Decrement Operator −−

Overload the pre-decrement operator −− to decrement weapons according to the following usage:

```
--ca1;
```

Note: Weapons can only be unloaded when on ground.

4.3.3 Output Stream Operator «

Ensure that your output stream operator « for `Aircraft` functions in line with calls of the following type:

```
cout << ca1;
```

The terminal output of this call should be exactly:

```
Identifier: CMBT1
Fuel Level: 89%
Health: 100%
Current Status: On Ground
Weapon Type: Missiles
```

```
Weapon Count: 11
Weapon Strength: 40
```

You may NOT define new output stream « operator overloading specifically for `CombatAircraft`.

4.4 `StealthAircraft` Operator Overloading

4.4.1 Cloak Status Operator !

Overload the logical NOT operator `!` to change cloak status according to the following usage:

```
StealthAircraft sa1("STLH1", 29, 34, AircraftStatus::Airborne, false); // Initialzed with cloak off

!sa1; // Turns cloak on
!sa1; // Turns cloak off
```

4.4.2 Output Stream Operator «

Ensure that your output stream operator « for `Aircraft` functions in line with calls of the following type:

```
cout << sa1;
```

The terminal output of this call should be exactly:

```
Identifier: STLH1
Fuel Level: 14%
Health: 24%
Current Status: Airborne
Cloak Status: Off
```

You may NOT define new output stream « operator overloading specifically for `StealthAircraft`.

4.5 `AbductorCraft` Operator Overloading

4.5.1 Abductee Ratio Operator >

Overload the greater than operator > to compare the abductee ratio of two AbductorCraft objects according to the following usage:

```
AbductorCraft ac1("ABDT1", 53, 98, AircraftStatus::Airborne, true, 3, 10); // Ratio = 3/10
AbdcutorCraft ac2("ABDT2", 42, 23, AicraftStatus::OnGround, false, 2, 5); // Ratio = 2/5

if (ac1 > ac2) {
    cout << "Greater\n";
}
else {
```

```
    cout << "Not Greater\n";
}
```

The terminal output of this call should be exactly:

```
Not Greater
```

The abductee ratio can be calculated using:

$$\text{Abdcutee ratio} = \frac{\text{abductees on board}}{\text{abductee capacity}}$$

4.5.2 Abduct Operator `+=`

Overload the addition assignment operator `+=` to simulate abductions according to the following usage:

```
ac1 += 7; // Abducting 7 small children.
```

Note: Abductions can only occur when airborne. The `+=` operator should not lose previous functionality when on ground.

4.5.3 Release Operator `-=`

Overload the subtraction assignment operator `-=` to simulate abductee release according to the following usage:

```
ac1 -= 3; // Dropping 3 small children.
```

Note: Abductee release can only occur when airborne. The `-=` operator should not lose previous functionality when on ground.

4.5.4 Merge Operator `&`

Overload the bitwise AND operator `&` to combine two abductor crafts according to the following criteria.

- Merge can only occur between two airborne abductor craft.
- The abductor craft with **higher abductee capacity** merges the other abductor craft with itself.
- The other abductor craft transfers **all** of its fuel, health, etc. to the abductor craft with higher abductee capacity.

Example usage:

```
ac1 & ac2; // ac1 (capacity 10) merges ac2 (capacity 5) with ↩
  itself
```

Notes:

- Think about what should happen to the other abductor craft i.e. ac2.
- What should happen in the case of `ac2 & ac1`?

4.5.5 Output Stream Operator «

Ensure that your output stream operator « for `Aircraft` functions in line with calls of the following type:

```
cout << ac1;
```

The terminal output of this call should be exactly:

```
Identifier: ABDT1
Fuel Level: 95%
Health: 100%
Current Status: Airborne
Cloak Status: On
Abductee Count: 9
Abductee Capacity: 15
```

You may NOT define new output stream « operator overloading specifically for `AbductorCraft`.

4.6 `GuardianCraft` Operator Overloading

4.6.1 Engagement Criteria Operator *=

Overload the multiplication assignment operator `*=` to check whether engagement criteria between a guardian craft and abductor craft is met:

- Both crafts must be airborne
- Guardian craft must have available weapons.
- Abductor craft must not be in stealth mode and must not have abductees on board.

Example usage:

```
GuardianCraft gc1("GRD1", 99, 84, AircraftStatus::Airborne, "Rockets", 6, 80, true, 0); // GuardianCraft with 6 rockets on board, weapon strength 80, cloak on, kill count = 0.
AbductorCraft ac1("ALIEN", 100, 60, AircraftStatus::Airborne, false, 0, 4); // AbductorCraft with health 60, cloak off, 0 abductees on board.

if (gc1 *= ac1) {
    cout << "Engagement Criteria Met\n";
}
else {
    cout << "Cannot engage\n";
}
```

The terminal output of this call should be exactly:

```
Engagement Criteria Met
```

Note: This operator is uni-directional.

4.6.2 Engagement Operator *

Overload the multiplication operator * to simulate fire from a guardian craft on an abductor craft.

- Each time * is used, one weapon is discharged
- Damage dealt to the abductor craft is according to weapon strength.
- Engagement can only occur if engagement criteria is met.

Example usage:

```
gc1 * ac1; // gc1 fires at ac1
```

Note: This operator is uni-directional.

4.6.3 Output Stream Operator «

Ensure that your output stream operator « for `Aircraft` functions in line with calls of the following type:

```
cout << gc1;
```

The terminal output of this call should be exactly:

```
Identifier: GRD1
Fuel Level: 99%
Health: 84%
Current Status: Airborne
Weapon Type: Rockets
Weapon Count: 5
Weapon Strength: 80
Cloak Status: On
Kill Count: 1
```

You may NOT define new output stream « operator overloading specifically for `GuardianCraft`.

5 Testing

To ensure the correctness and robustness of your implementation, testing will be conducted using the provided test file `Test_ACS_61.cpp`. This file is used to run a series of predefined test cases that will evaluate various functionalities of your classes.

All testing will be done on Docker.

You may compile the executable file using:

```
g++ -std=c++11 -o a ACS_61.cpp Test_ACS_61.cpp
```

and run via

```
./a
```

You only need to compile this file once, and you do not need to manually re-compile test individual test files after making changes to your code. Simply select the desired index from the list and your files will be recompiled for you.

Note: Be sure to **define all methods of a class** in your code before running the test cases for a particular portion i.e. your test cases for `Aircraft` will not compile unless all constructors, getters, `land()`, and `takeoff()` have been defined.

Please find answers to FAQs here.

Good luck and Start Early!